Unit 2 - Computer Systems
Assignment 7 - Managing Upgrades (Part 1)

Introduction

Scenario/Vocational Context: You are a Junior Technician in Jersey Bank and have been asked to review the reasons for upgrades within the Bank. You have been asked to prepare a document.

Task 1

(U29:P6) - Explain the importance of the user acceptance process

Task description: *The organisation needs to know that a newly installed or updated system will function correctly.*

Using your system as an example (Windows 8.1 to Windows 10), explain what a user acceptance process would involve

Basically, user acceptance is the process of reviewing a product or system, which is either carried out before sending out a product or when carrying out an upgrade on computer systems.

What I mean by this is that it basically is a public beta test, which allows the public to find previously unknown faults with the product so it can be addressed by the manufacturer or to see whether it is worth to upgrade the system to that specific product or system.

With the User Acceptance process, there are different steps taken for the UAT. Which can typically involve one or more of the following: User Acceptance Test (UAT) Planning, Designing UA Test Cases, Selecting a Team that would execute the (UAT) Test Cases, Executing Test Cases, Documenting the Defects found during UAT, Resolving the issues/Bug Fixing and a Sign Off.

User Acceptance Test (UAT) Planning

For a manager who overviews the installation/upgrade process, they would need to make sure that the operating system will have applications that the worker would feel as comfortable as the previous system, so planning would be needed to carry out a User Acceptance Test.

To carry out a UAT, the manager would have to make a plan of everything that is going to be done within the UAT. The plan can consist of who is going to be in the team, what tests they are going to carry out, what the team will document and how to debate if the operating system will pass the test or not.

For instance, with the installation of Windows 10 from Windows 8.1 on a computer system, it would need to have a user acceptance process. The reason for this is that, once software has been fully upgraded or installed, it still needs to match the user's requirement.

Designing UA Test Cases

With the previous planning, the manager could start designing the UA Test cases. The way this is done is mostly due to the way the manager will design the UAT cases. Which means that this can be done in different ways.

For instance, they can look at how the operating system was being used before in terms of what shortcuts there may have been or what user interface the previous version had. This could be done using a selected team that would execute the UAT test cases.

However, the test cases would have to consist of methods on getting the most beneficial result for the documentation and sign off, so it could be determined whether the operating system or any other product can be used in a business or not.

Selecting a Team that would execute the (UAT) Test Cases

Usually, the manager would get members of staff as a testing team, and let them tell the manager what is needed in order to make their job on a computer system much more pleasing and efficient, by checking whether the system has no defects and fits the user requirement comfortably.

With the team's feedback, the system can be decided whether it fits their need as well as let them find any malfunctions within the new system, so it could be used to inform the manufacturer to perform patches.

Executing Test Cases

One method of the execution of the test cases for the user acceptance process, would be to give instructions to the team on how to use the software, to get feedback on whether they would find it better or worse, which would be done using documents. These formal documents will contain a manual for how to use certain aspects for the software, so they can get a good idea on how to navigate and use the new OS for testing.

If an error was to be found within the user acceptance process, the team would have documents such as: the installation reports (which can be handed over to IT management or further), contact details of the installer, the company from which the software is and other technical support.

After the testing for any bugs or malfunctions, it will now be up to the user if they see the operating system being something they keep or something they wish to revert back. For this, they will need to have a hands on experience over time to be able to judge for themselves if they have made the right choice of upgrading the OS or not.

They will also need to test other software within the OS to see if they too run fine without any malfunctions or crashes. If all is good then they will need to sign for it being done efficiently and that they wish to carry on with the upgrade until further notice.

However, if for whatever reason the software upgrade was a problem due to something that the members of the testing team didn't have any responsibility to the cause of difficulties. Then, as part of customer trust, they would need go about perform a reverse effect of uninstalling the software or going back to how the system was before the installation.

Documenting the Defects found during UAT

After all the test cases have been executed, the process that the testing team has carried out would have to be documented, having written down what problems have occurred and other forms of defects.

For example, documenting the defects that the testing team has found on the operating system, it gives them the ability to collect all the data together, finding different bits of faults that have been found by the team and even which fault was the most common.

Resolving the issues/Bug Fixing

With this information, the company can contact the manufacturer on any defects that has been found and the information on how the defect occurred. In which the manufacturer of the system would try and resolve the issues by carrying out bug fixes and other forms of patches.

For example, with the documented defects on the operating system, the test manager can inform the manufacturer on these defects. Which the manufacturer would then carry out fixes for the specific defects that has been found by the UAT team.

Sign off

Lastly, (after the planning, testing, documentation and fixes) the upgrade would have to be signed off. Which would have the final say on whether the upgrade is worthwhile or that the previous version is much better to the version that has been tested.

For example, if the operating system passed the test with minor faults, with those faults being addressed, the team would accept the upgrade for the company. Which would be done by the

team signing a document which confirms that the operating system is in good working condition and is suitable for the upgrade.

Explain, with a detailed example, what would happen if there was no process to confirm the functionality of the system (Windows 10) and if the users were unable to use the upgraded system

If there wasn't a form of process to confirm the functionality of a system, it would be highly likely that the system would have faults that was unknown by the manufacturer and the company using the new upgrade. This means that the system would have poor user friendliness, causing it to have a bad user reviews and poor work output.

This would have a massive impact on the system, since the purpose of an operating system is to access the functionality of the computer by being user friendly. With the operating system having many possible faults, it would mean that the user friendliness would be instantly removed.

So, removing each step from the user acceptance test can each have its own impact, some being a lot more significant than others. The reason being is because each step of the user acceptance test can prove to be important for the testing of the operating system.

User Acceptance Test (UAT) Planning

Without the UAT planning, the whole of the user acceptance process will be a significant mess and would be very hard to maintain, causing it to be an obsolete test due to the lack of organisation.

What I mean by this is that, if the manager doesn't choose what is going to be in the test and how it's going to be done, the test would become a free for all without any form of recording or knowing what they are supposed to test and document.

For example, if there was no plan prior to a test plan and the manager decides to get a random selection of staff members to check for the functionality of the OS, it would be very disorganised. The reason being that it would be unlikely that they would record the correct or sufficient amount of data which contains the fault within the new OS, which means the test wouldn't be very useful.

Overall, without any form of recorded testing and documentation, it would be very unlikely that the test would cover everything and have a refined verdict on whether the OS should be used by the company. This reason is why the UAT planning is really important to have before the actual UAT.

Designing UA Test Cases

Without the designing of the UA test cases, the whole of the user acceptance process would yet again be a mess and would be very hard for the testers to follow, causing it to be difficult for the testers to know what to do.

What I mean by this is that, if the testers don't know what they are going to be doing in the test, the test would become very hard to know what they are supposed to record, making it very hard for the test to be done.

For example, if there was no UA test cases on a test plan and the testers carry out random functionality tests for the functionality of the OS, it would yet again be very disorganised. The reason being that, yet again, it would be unlikely that they would record the correct or sufficient amount of data which contains the fault within the new OS, which also means the test wouldn't be very useful.

Overall, without the UA test cases, it would be very unlikely that the testing team would yet again cover everything and have a refined verdict on whether the OS should be used by the company. This reason is why the UA test cases is really important to have for the UAT.

Selecting a Team that would execute the (UAT) Test Cases

Without a team that would execute the UAT test cases, the user acceptance process would be very difficult to follow by a single person or by random staff members, causing it to be difficult for the testing of the OS.

What I mean by this is that, if there isn't a team for the testing it would be extremely difficult to find a lot of faults within the operating system. Meaning that the test would contain very little documented information, which means the test wouldn't be very useful.

For example, if there was no team on a test plan to carry out the tests for the functionality of an OS, it would again be very disorganised. The reason being that, again, it would be unlikely that the sufficient amount of data would be recorded, which again means the test wouldn't be very useful.

Overall, without a team that would execute the UAT, it would be very unlikely that there would be a good verdict on whether the OS should be used by the company. This reason is why a team that would execute the UAT is really important to have for the UAT.

Executing Test Cases

Without the execution of test cases, the test wouldn't occur in the first place. The reason being that the execution of test cases is what makes the entire testing of the UAT, meaning that this is the most significant part of the UAT.

What I mean by this is that, the execution of test cases is literally the team carrying out the UAT. Basically, without the execution for the test cases, there wouldn't be a User Acceptance Process due to it revolving around the execution of the test cases.

For example, if a manager planned, designed the test cases and got a team together and then does not carry out the execution of the test cases. It would mean that the previous design and planning would be useless due to the fact that it hasn't been carried out by the team.

Overall, the execution of the test cases is the most vital part of the UAT due to it being the main part of the UAT. This is important because it would make the other processes useless without an execution of the actual test cases, meaning that this is needed, while others can be left out and still carry out a somewhat useful user acceptance test.

Documenting the Defects found during UAT

Without the documentation of the defects found during UAT, the user acceptance test would be very difficult to gather information about the faults and errors that have been found, causing it to be difficult for a report to send to the manufacturer to carry out bug fixes.

What I mean by this is that, if there isn't documentation for the testing it would be extremely difficult to find a lot of faults within the operating system. Meaning that the test would contain very little documented information, which means the test wouldn't be very useful.

For example, if there was no documentation from the UAT, it would again be very disorganised and would be difficult to send the issues to the manufacturer. The reason being that it would be unlikely that the sufficient amount of data would be recorded, which again means the test wouldn't be very useful.

Overall, without documentation that would execute the UAT, it would be difficult to send the information about faults and errors towards the manufacturer of the operating system, making the bug fixing and resolvement of other issues almost impossible to do. This reason is why a team that would execute the UAT is really important to have for the UAT.

Resolving the issues/Bug Fixing

Without the resolving of the issues and bug fixing, the OS wouldn't be able to be accepted due to the errors that are still present. The reason being that the errors that would most likely cause the operating system to malfunction, crash or to be very unresponsive, causing it to be very unreliable.

What I mean by this is that, by fixing of the errors it would make the operating system usable by the company, and without the repair of these errors, it could mean that it would be practically unusable.

For example, if a manager planned, designed the test cases, got a team together, carries out the execution of the test cases and doesn't have any documentation of faults that have been found within the testing, it would mean that the previous planning and execution would have a chance on being useless due to the fact that it has a high risk of the errors being significant to the usage of the operating system.

Overall, having documentation of the defects that have been logged in the test is important to have because it is solid proof that the test has been carried out and that the errors found has been logged to be sent to the manufacturer of the operating system. This is important because it has a chance to make the other processes useless, meaning that this is needed for evidence.

Sign off

Without the sign off, the OS again wouldn't be able to be accepted due to the lack of physical acceptance from the team. The reason being that the team will need to sign a sing off in order for the operating system to have proof that it has been tested for use.

What I mean by this is that, by having solid proof that the operating system has been tested and accepted/declined for use, it gives the verification that the operating system has been tested properly. Meaning that the sign off would be proof the the company and giving them the ability to use the operating system without a lot of worrying.

For example, if a manager planned, designed the test cases, got a team together, carries out the execution of the test cases, has documentation but doesn't have a sign off of the test, it would mean that the previous testing would be useless due to the fact that there is no proof that the operating system has not been tested, meaning that the company cannot use it due to lack of actual proof.

Overall, having a sign off of the tests is important to have because it is solid proof that the test has been carried out and that the errors found has been addressed and that the rest of the operating system is good for business use. This is important because, without this, the other processes would be useless due to lack of evidence, meaning that this is also needed for evidence.

(U29:M1) - Explain the advantages and potential disadvantages of installation or upgrade of new software

Task description: *Your manager would like you to create a document for the Finance Department members explaining the advantages and disadvantages of upgrading or installation.*

For each installation/upgrade you need to consider 3 advantages (benefits to the user and to the Company) and 3 disadvantages (What can go wrong and/or what would happen if the upgrade/installation is not performed)

2.7 Task 2 Planning Grid Upgrades	Explanation of the Advantages/Benefits of the Upgrade	Explanation of Disadvantages/problems of the Upgrade.
Upgrade from Windows 8.1 to Windows 10	Firstly, by upgrading to Windows 10, a company gains advantages such as the lack of errors that would be found in the previous operating system. New operating systems always has improved the bugs that have been present on the past version of the OS, which can cause the system to crash, freeze or malfunction. Which would save the company time as staff are able to use the operating system without it crashing. Secondly, another advantage of upgrading is, faster application execution. In simpler terms, this makes the software and the OS open and function faster. This will also save the company some time as the staff will not require to wait an excessive amount of time for an application to open or for the OS to load. Lastly, newer operating systems can offer additional functionality and extra features, which can help the users to get work done faster. Additional features can also make certain parts of the operating	Firstly, the new support from the manufacturer can be a big disadvantage to a company due to the fact the operating system is massively unknown in terms of errors. For instance, if a business has used an operating system for a long period of time, the software previously used could cancel the support for the newer operating system, or very little support. Which means that the current bugs on that software used on the newer operating system will take a long time to be fixed and would be more vulnerable. Secondly, another disadvantage is that it could also mean that the package would not be compatible with the previously used software or hardware. Therefore, the company would require to look for a new software or an upgrade which would increase the cost budget significantly. Lastly, it is not always ensured that new software will have no bugs or other complications just because it is new. This could be a huge disadvantage to a company that decides to upgrade from old software to new, to only find out that the new software is more buggy than the old one.

	system more convenient to use than in the previous version of the operating system. Overall, additional features can make employees work more efficiently and therefore earning more money for the company.	The bugs would eventually get fixed, however it would take some time for the updates to be relapsed, therefore potentially costing the company more money.
Upgrade from Office 2013 to 2016	Firstly, by upgrading to office 2017 from 2013, a company gains advantages such as the lack of errors that would be found in the previous operating system. Newer software always has improved the bugs that have been present on the past version of the software, which can cause the software to crash, freeze or malfunction. Which would save the company time as staff are able to use the software without it crashing, losing vital work. Secondly, another advantage of upgrading is, faster application execution. In simpler terms, this makes the software and the features within the software open and function faster, which means that more work is produced. This will also save the company some time as the staff will not require to wait an excessive amount of time for an application to load. Lastly, the new Office 2016 can offer additional functionality and extra features, which can help the users to get work done at a much faster pace and at a better quality. Additional features can also make certain parts of the software more	Firstly, the new support from the manufacturer can be a big disadvantage to a company due to the fact the software is massively unknown in terms of faults. For instance, if a business has used the older version of Office for a long period of time, the software previously used could cancel the support for the newer operating system, or very little support. Which means that the current bugs on a older software will take a long time to be fixed and that the newer Office would be more vulnerable to faults. Secondly, another disadvantage is that it could also mean that the newer Office 2016 would not be compatible with older software or hardware. Therefore, the company would require to look for a new software or an upgrade which would increase the cost budget significantly. Lastly, it is not always ensured that new software will have no bugs or other complications just because it is new. This could be a huge disadvantage to a company that decides to upgrade from old software to new, to only find out that the new software is more buggy than the old one. The bugs would eventually get fixed, however it would take some time for the updates to be relapsed, therefore

	convenient to use than in the previous version of the operating system. Basically, additional features can make employees work more efficiently and therefore earning profit for the company. Overall, the benefits to the Office 2016 software is similar to the benefits of the OS, due to the fact that newer software in any form usually is better performing and has more functionality.	potentially costing the company more money. Overall, the drawbacks to the Office 2016 software are also similar to the drawbacks of the OS, due to the fact that newer software in any form usually is better performing and has more functionality.
Install Norton Anti-Virus rather than use Windows Defender	Firstly, by upgrading to Norton Antivirus from Windows defender, a company gains advantages such as the lack of errors that would be found in the previous operating system. The improved anti-virus always has improved the efficiency and speed when compared the the default Windows Defender, which means the software can detect and remove malware off a PC with more efficiency. Which would save the company time as staff are able to use the anti virus without it crashing and at a faster detection speed. Secondly, another advantage of upgrading is, the automated malware detection. In simpler terms, this means that the Norton antivirus can detect viruses and other forms of threats automatically. This will also save the company some time as the staff will not require to manually scan and wait an excessive amount of time for the application to finish scanning.	Firstly, the new third party software from the manufacturer can be a big disadvantage to a company due to the fact the software is not a default program in the Windows operating system, which can be a disadvantage. For instance, having a third party software means that it would have to be updated separately to the OS, meaning that a mass update would be needed on every computer system of the company, Which means that it can be time consuming if there were errors such as file corruptions or crashes, meaning that the company would need to stop the work output in order to address these issues. Secondly, another disadvantage is that it could also mean that the package would not be compatible with newer operating software or hardware. Therefore, the company would require to look for a new operating software or an upgrade which would increase the cost budget significantly. Lastly, it is not always ensured that new updates for the software will have no bugs or other complications just because it is new.

	Lastly, Norton antivirus has many updates and extra features, which means that the antivirus software will always be on time with newer threats to the computer system, reducing them. Additional updates and features can also make certain parts of the software more convenient to use when compared to the Windows defender. Overall, additional updates and features can make the software much more reliable and efficiently due to the fixes and improvements, reducing the threat that the business has..	This could be a huge disadvantage to a company that decides to upgrade from old update to new, to only find out that the new update is more buggy than the old one. The bugs would eventually get fixed, however it would take some time for the updates to be relapsed, therefore potentially costing the company more money. Overall, the drawbacks to the Norton antivirus software is similar to the drawbacks of the previous OS, due to the fact that software has similar drawbacks due to the fact that they are all stored within the PC the same way, and that maintenance is needed for each bit of software, no matter the purpose.